

SAMEER KARUR

Data Analyst & Technical Consultant | Applied AI & Machine Learning Practitioner

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PROFESSIONAL SUMMARY

Data Analyst & Technical Consultant with 2.5+ years of experience transforming complex enterprise datasets into actionable business intelligence, automated analytical pipelines, and mission-critical enterprise systems. Proven track record spanning high-scale consumer e-commerce fulfillment (**Flipkart**) and global life sciences & pharmacovigilance technology (**ArisGlobal**). Advanced hands-on expertise in SQL, Python, and Tableau, complemented by specialized rigor in Generative AI (RAG, LangChain, Vector Databases), Deep Learning (TensorFlow, Computer Vision), and Predictive Machine Learning through **E&ICT; Academy, IIT Kanpur**.

TECHNICAL SKILLS & COMPETENCIES

Analytics & BI:	Advanced SQL (Window Functions, CTEs, Complex Joins), Tableau Dashboard Design, Exploratory Data Analysis (EDA), KPI Tracking, Root Cause Analysis, Statistical Hypothesis Testing, Supply Chain & Operations Analytics
Programming & DB:	Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), JavaScript, MySQL, Relational Database Management (RDBMS), Data Extraction & Automated ETL Pipelines
Machine Learning:	Supervised & Unsupervised Learning, XGBoost, Random Forest, Logistic Regression, K-Means Clustering, SMOTE Imbalance Handling, Time-Series Demand Forecasting
Deep Learning & CV:	TensorFlow, Keras, Convolutional Neural Networks (CNNs), Transfer Learning (MobileNetV2), Computer Vision (OpenCV), Multi-Class Classification & Object Localization
Generative AI:	Retrieval-Augmented Generation (RAG), LangChain, ChromaDB Vector Database, OpenAI APIs, Prompt Engineering, Gradio Web UI Deployment
Cloud & Developer Tools:	Microsoft Azure (AZ-900 Certified), Git, GitHub, Jupyter Notebook, VS Code, Advanced Excel

PROFESSIONAL EXPERIENCE

Consultant — ArisGlobal	Bengaluru, India Apr 2026 – Present
Regulatory Data Engineering: Configure business validation rules, automated regulatory workflow logic, and data verification scripts using SQL and JavaScript for enterprise Pharmacovigilance (PV) safety platforms. Data Integrity & Compliance: Query, audit, and validate multi-tiered clinical and adverse event (AE) databases to guarantee data completeness and strict adherence to global health regulatory standards (FDA, EMA). Automation & ETL Pipelines: Develop Python and SQL automation scripts for structured data extraction, schema validation, and reporting consistency across high-throughput client safety systems. Stakeholder Consulting: Interface directly with cross-functional technical teams and client domain leads to troubleshoot mission-critical data anomalies, optimize data flows, and accelerate case processing throughput.	

Data Analyst — Flipkart (Payroll: RLabs Enterprise Services)	Bengaluru, India Jan 2024 – Apr 2026
Executive BI & Dashboards: Architected and maintained executive Tableau dashboards monitoring 10+ operational KPIs (SLA compliance, fulfillment throughput, breach rates, delivery TAT) consumed daily by business and city operations leadership. Big Data SQL Analytics: Authored high-performance SQL queries (window functions, recursive CTEs, complex joins) across multi-million row transactional databases to conduct root-cause investigations on operational bottlenecks. Process Automation: Built automated data extraction, cleaning, and reporting pipelines in Python (Pandas, NumPy), slashing recurring manual reporting turnaround time by over 70% and saving 10+ engineering hours weekly. Cross-Functional Operations Impact: Partnered with warehouse, supply chain, and logistics leadership to translate ambiguous fulfillment delays into structured metrics, directly informing shift scheduling, capacity planning, and resource allocation. Data Quality Governance: Built automated discrepancy audits reconciling warehouse log systems with downstream reporting tables, ensuring 99%+ reporting reliability.	

FEATURED AI & MACHINE LEARNING PROJECTS

Python, LangChain, ChromaDB, OpenAI Embeddings, GPT-3.5 Turbo, Gradio

AI-Powered Enterprise HR Assistant (RAG System & Gradio UI)

- Architected a production-ready conversational RAG assistant grounded in official multi-page PDF policies to resolve employee HR inquiries in natural language.
- Chunked documents with RecursiveCharacterTextSplitter and generated vector embeddings persisted in ChromaDB with cosine similarity retrieval.
- Enforced strict contextual grounding prompts that eliminated hallucinations, deploying an interactive Gradio web chat interface.

TensorFlow, Keras, MobileNetV2, OpenCV, Scikit-learn, Pandas, Seaborn

Autonomous Driving: Vehicle Perception & Tesla Crash Analytics

- Trained a transfer-learning MobileNetV2 crop classifier on 5,600+ road scene images with 17,900+ bounding boxes, achieving **89.9% coarse classification accuracy** across multiple vehicle classes.
- Conducted deep safety analytics on 293 fatal Tesla crash records (352 deaths), analyzing victim roles, collision mechanics, model distributions, and empirical verification of Autopilot-involved incidents.

Python, XGBoost, Random Forest, Scikit-learn, Pandas, Time-Series Modeling

Multi-Restaurant Demand & Sales Forecasting System

- Analyzed 109,600 item-day transaction records across 6 restaurants and 100 menu items to forecast daily demand.
- Engineered time-series calendar features (seasonality, day-of-week, quarter, revenue vs. volume velocity).
- Built and evaluated Linear Regression, Random Forest, and XGBoost; champion XGBoost achieved **RMSE 57.90 (R² 0.948)** on a 6-month test holdout, generating an accurate full-year forward demand forecast.

TensorFlow, Keras, MobileNetV2, Scikit-surprise (SVD), OpenCV, Pandas

Preserving Cultural Heritage: Monument Classification & Recommender

- Developed a dual-engine AI platform combining Deep Learning computer vision with an intelligent tourism recommendation system.
- Built a MobileNetV2 classifier across 10 global heritage landmarks, achieving **86.5% test accuracy** with data augmentation.
- Engineered a personalized collaborative filtering recommender with SVD and cosine similarity, serving top-N customized itineraries.

Python, OpenAI DALL-E / Images API, GPT-4 / ChatGPT, Gradio UI, PIL

Generative AI Marketing Campaign Asset & Pitch Generator

- Built an end-to-end GenAI application that takes natural-language brand concepts and generates professional marketing pitch copy and 4K campaign imagery.
- Designed chained system prompts combining visual prompt expansion with structured marketing copy generation.
- Deployed an intuitive Gradio web UI allowing non-technical marketing teams to prototype multi-channel ad concepts in seconds.

Python, Scikit-learn, Random Forest, SMOTE, K-Means Clustering, Seaborn

Predictive Employee Turnover Analytics & Retention Modeling

- Modeled attrition patterns across 15,000 employee profiles; clustered key dissatisfaction personas via K-Means.
- Resolved severe class imbalance via SMOTE resampling; achieved **97.2% accuracy and 0.98 ROC-AUC** with Random Forest, segmenting staff into prioritized retention risk tiers for targeted HR action.

EDUCATION & CERTIFICATIONS

Professional Certificate in Generative AI and Machine Learning E&ICT; Academy, IIT Kanpur (Dec 2025 – Nov 2026)

*Top Capstone Delivery & Distinction
10+ Real-World AI Implementations*

**Bachelor of Engineering (B.E.) in Electronics & Communication
Vishwanathrao Deshpande Institute of Technology, Karnataka (Graduated: Aug 2022)**

*First Class with Distinction
Technical Foundations*

Professional Certifications: Microsoft Certified: Azure Fundamentals (AZ-900) | SQL for Business Analytics | Python for Data Analysis | Exploratory Data Analysis with Tableau